

SpaceY | Data Science Team

SpaceX Falcon 9 Launch Analisys and Prediction

Presented by Gabriel Gálvez

October 2026

Gabriel Galvez

Project Overview

1. Executive Summary.
2. Methodology.
3. Conclusions and Key Findings.
4. Appendix.

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Executive Summary

Methods.

- Data collection through the SpaceX API and web scraping.
- Data cleaning and exploratory analysis using Python, Pandas, visualization libraries, and SQL.
- Interactive visual analytics with Folium and Plotly Dash.
- Predictive modeling using Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.

Key Findings.

- Launch outcomes vary across launch sites, payload ranges, and booster versions.
- The Decision Tree achieved the highest test accuracy among the evaluated models: 94.44% (17 out of 18 test samples).
- The small test set limits the certainty with which this result can be generalized.

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Introduction

SpaceX has reduced launch costs by developing reusable rocket technology. Recovering and reusing the Falcon 9 first stage is an important part of this approach.

Project Objective.

Analyze publicly available Falcon 9 launch data to identify patterns in launch outcomes and build a machine learning model to predict first-stage landing success.

Business Context.

SpaceY is a fictional competitor seeking data-driven insights to support launch-cost estimates and strategic decision-making.

Scope.

- Explore launch sites, payload masses, orbit types, and landing outcomes.
- Investigate historical patterns through data visualization and SQL.
- Present findings through interactive visual analytics.
- Compare classification models and evaluate their predictive performance.

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Methodology



Methodology Index

1. SpaceX API Data Collection.
2. Web Scraping.
3. Data Wrangling.
4. Exploratory Data Analysis (EDA).
5. EDA with SQL.
6. Interactive Visual Analytics.
   - Folium Maps.
   - Plotly Dash Dashboard.
7. Predictive Analysis.

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SpaceX API Data Collection

Data Collection Process.

1. Send HTTP requests to the API endpoints.
2. Retrieve launch, rocket, payload, launch site, and landing outcome data.
3. Parse the JSON responses into structured Python data.
4. Transform the records into a Pandas DataFrame.
5. Save the dataset for further analysis.

Data Handling.

Live API access was unavailable during the lab, so the supplied JSON dataset was used to continue the analysis.

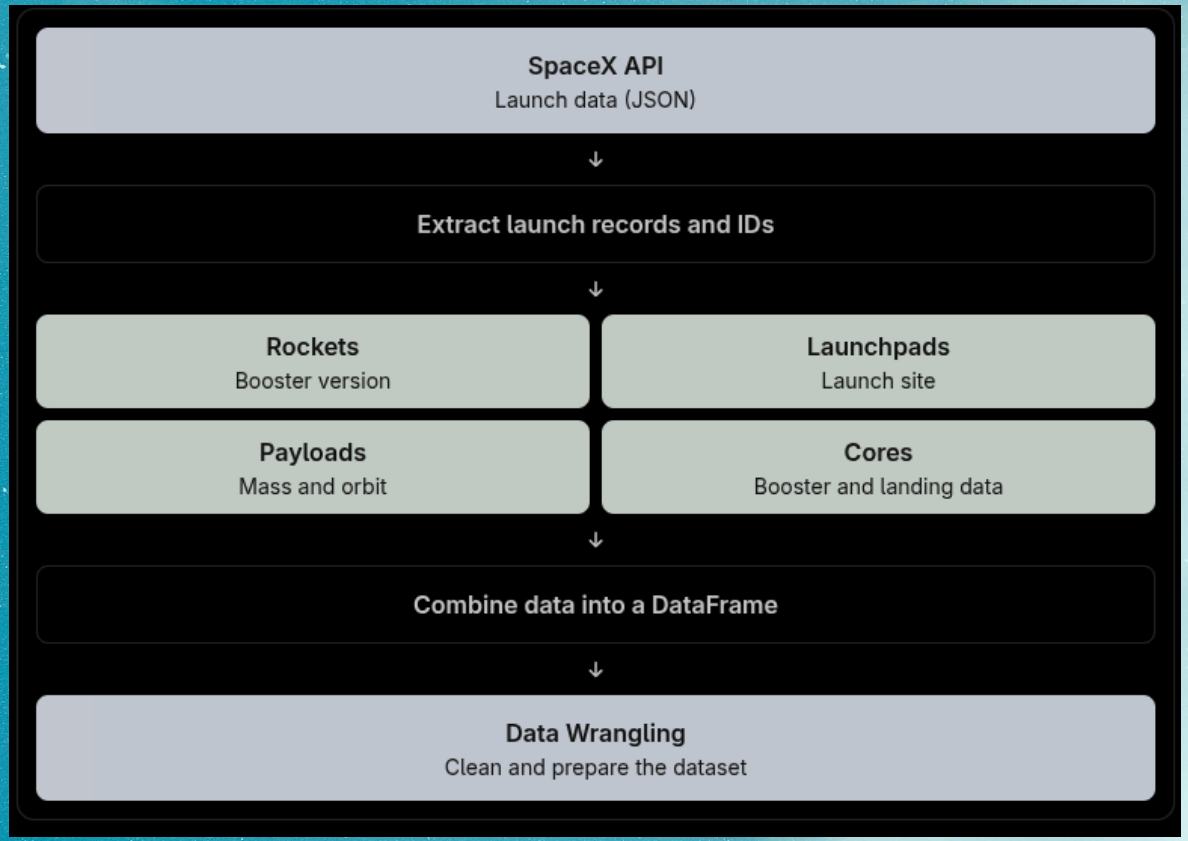
Output.

Structured launch data for exploratory analysis and predictive modeling.

LAB

0101_jupyter_labs_spacex_data_collection_api_final.ipynb

SpaceX API Data Collection



LAB

Web Scraping

Data Source.

Historical Falcon 9 and Falcon Heavy launch records from Wikipedia.

Scraping Process.

1. Retrieve the web page using Python Requests.
2. Parse the HTML content with BeautifulSoup.
3. Identify the launch-history tables.
4. Extract and transform the table data using Pandas.
5. Export the structured records to CSV for further analysis.

Output.

A structured dataset of historical launch records to complement the API data.

Source.

Wikipedia: List of Falcon 9 and Falcon Heavy launches.

LAB

0102_jupyter_labs_webscraping_final.ipynb

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Data Wrangling

Data Preparation.

The collected launch records were cleaned and transformed to prepare the dataset for exploratory analysis and machine learning.

Main Steps.

1. Handle missing values and inconsistent records.
2. Convert dates and numerical fields to appropriate data types.
3. Select relevant launch, payload, orbit, and landing outcome features.
4. Create the binary landing outcome classification (Class).
5. Prepare the final dataset for visualization and predictive modeling.

Output.

A structured dataset with consistent features and a binary target variable for modeling first-stage landing success.

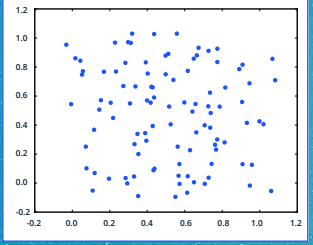
LAB

0103_jupyter_labs_spacex_data_wrangling_final.ipynb

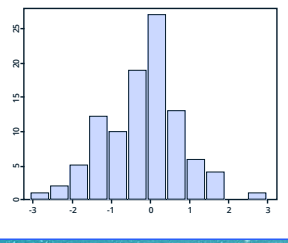
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EDA Visualization

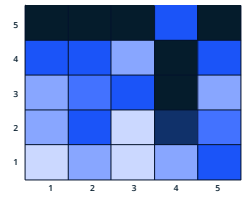
Scatterplot



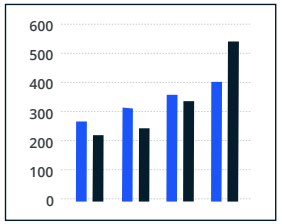
Histogram



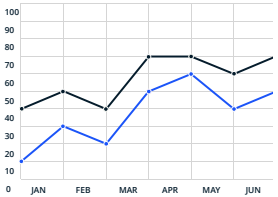
Heatmap



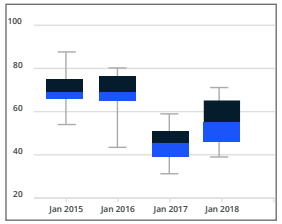
Bar Chart



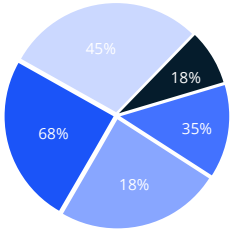
Line Chart



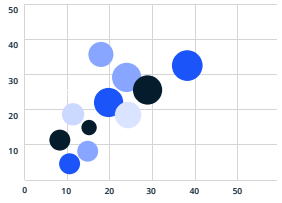
Boxplot



Pie Chart



Bubble Chart



LeewayHertz

LAB

EDA with Data Visualization

Objective.

Explore relationships between launch characteristics and first-stage landing outcomes.

Visualization Methods.

- Compare flight number and launch site.
- Analyze payload mass by launch site and orbit.
- Examine landing success across orbit categories.
- Investigate annual landing success trends.
- Encode categorical features for subsequent machine learning.

Tools.

Python, Pandas, Matplotlib and Seaborn.

Purpose.

Identify patterns and relationships that may help explain landing outcomes and guide predictive modeling.

LAB

0202_jupiter_labs_eda_dataviz_final.ipynb

EDA Visualization : Results

1. Launch Characteristics.

a) Flight Number vs. Launch Site.
b) Payload Mass vs. Launch Site.
c) Succes Rate vs. Orbit Type.

2. Orbit and Payload Analysis.

d) Flight Number vs. Orbit Type.
e) Payload Mass vs. Orbit Type.
f) Annual Landing Success Trend.

The visual analysis explores launch patterns, payload characteristics, orbit categories, and changes in landing success over time.

The texts describe what each chart helps us analyze.

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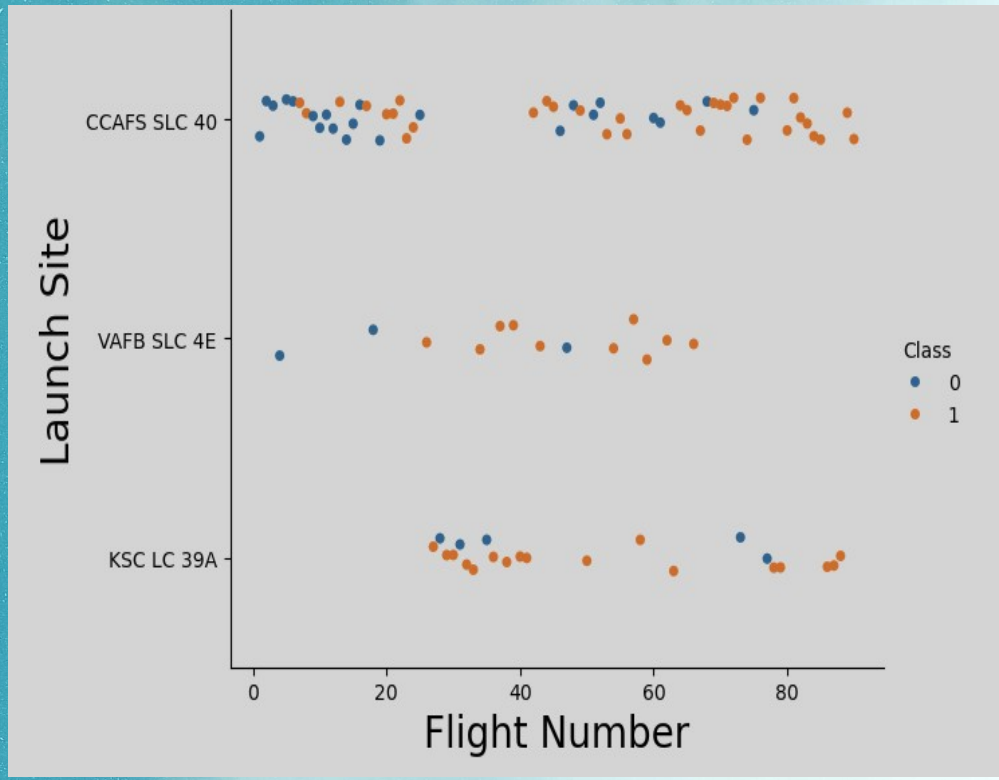
EDA Vis: *Launch Characteristics*

1. a. Flight Number vs. Launch Site.

The distribution of launches varies by launch site.

As the flight number increases, the outcomes across launch locations can be compared to explore the relationship between accumulated launch experience and mission success.

GRAPH

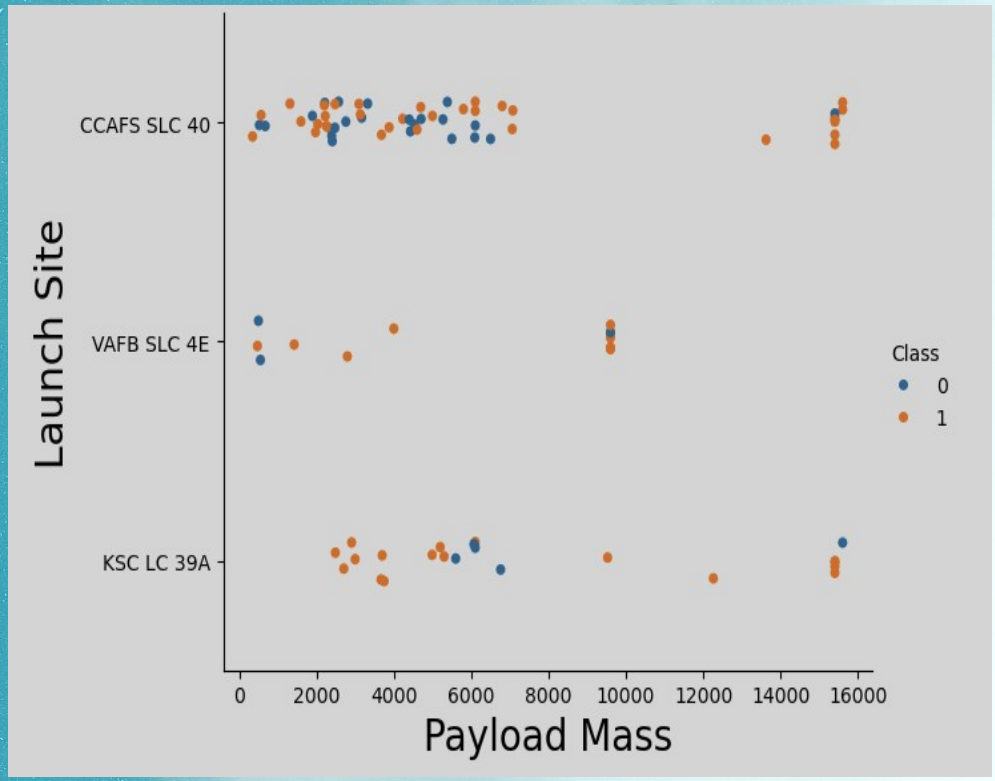


EDA Vis: *Launch Characteristics*

1. b. Payload Mass vs. Launch Site.

Payload mass allows us to compare the types of missions conducted from each launch site.

The chart shows the distribution of payload masses and launch outcomes, helping identify whether certain sites handle heavier payloads or display different success patterns.



GRAPH

EDA Vis: *Launch Characteristics*

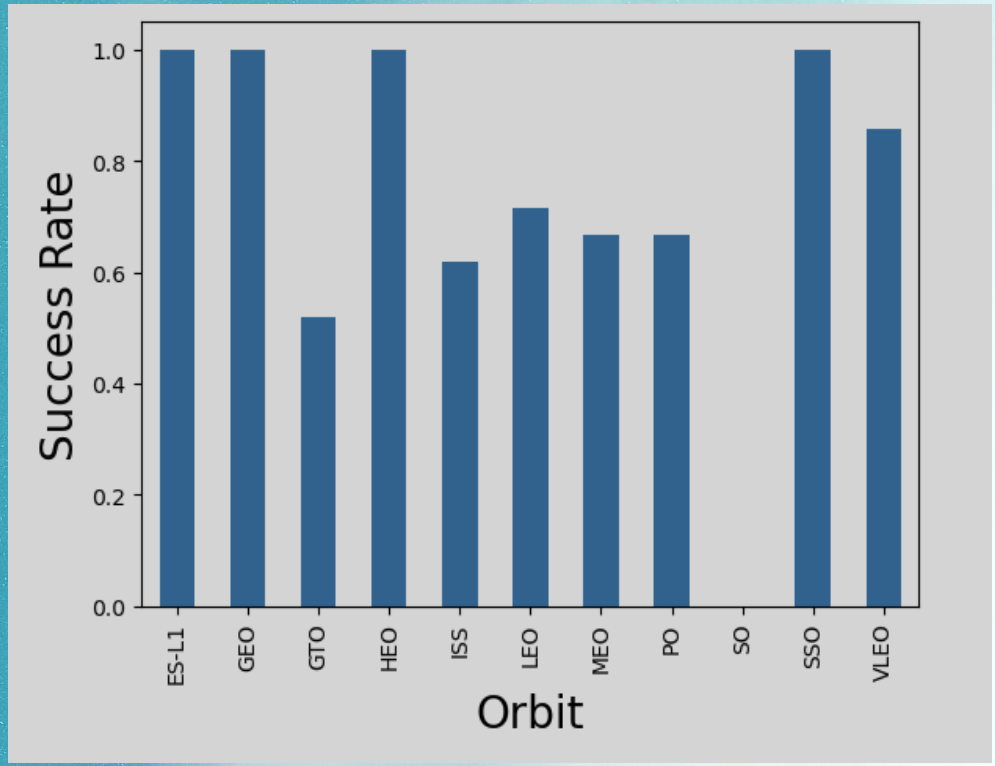
1. c. Success Rate vs. Orbit Type.

The success rate varies across orbit types.

This comparison helps identify which orbit types have higher or lower success rates.

The number of launches in each category should also be considered when interpreting the results.

GRAPH

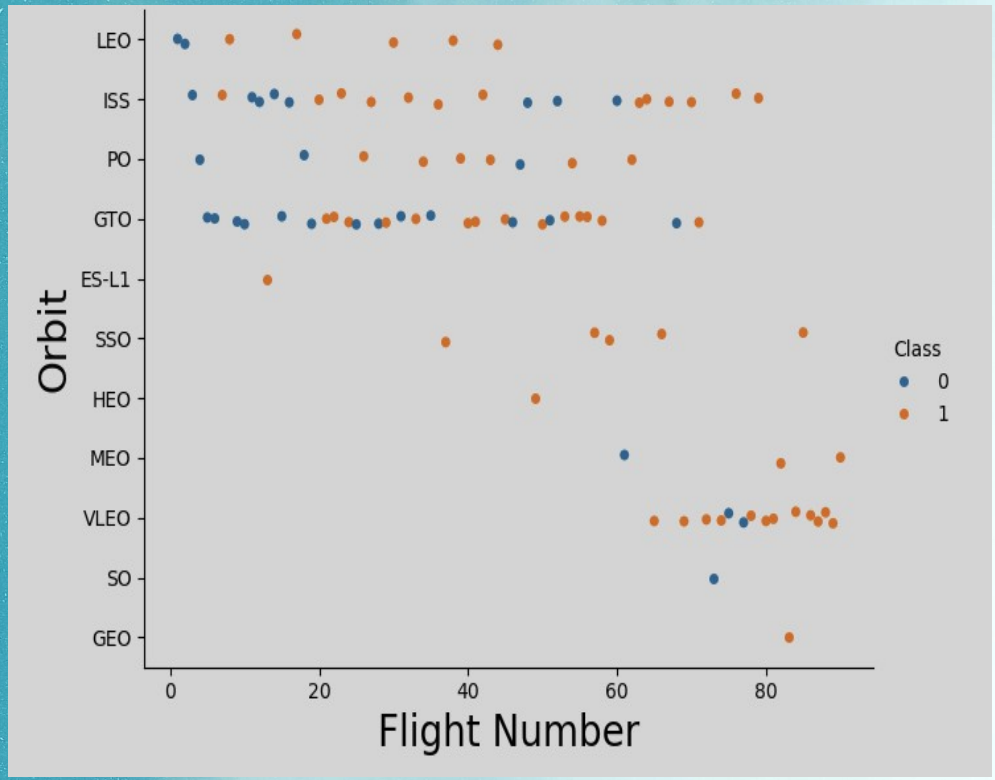


EDA Vis: *Orbit and Payload Analysis*

2. d. Flight Number vs. Orbit Type.

The chart shows how launches are distributed across different orbit types throughout the flight sequence.

It helps identify which orbit types appear at different stages and how successful and unsuccessful outcomes are distributed.



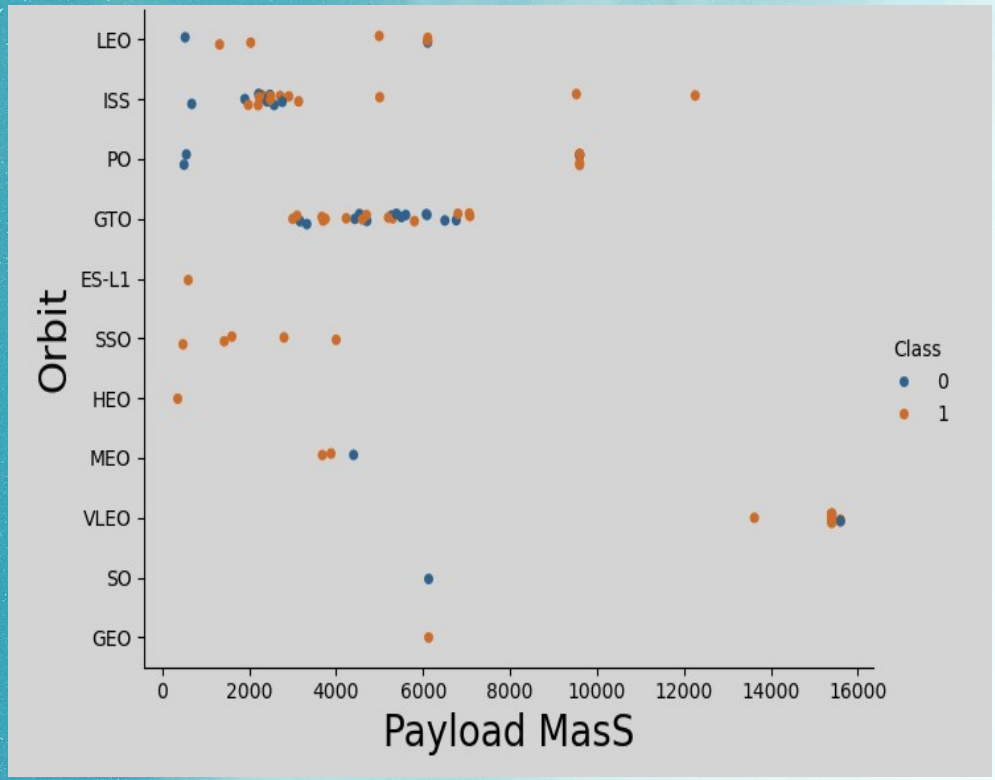
GRAPH

EDA Vis: *Orbit and Payload Analysis*

2. e. Payload Mass vs. Orbit Type.

Payload mass varies across destination orbit types.

Comparing the results helps explore whether certain payload ranges are associated with specific orbits and whether payload mass appears to be related to mission success.



GRAPH

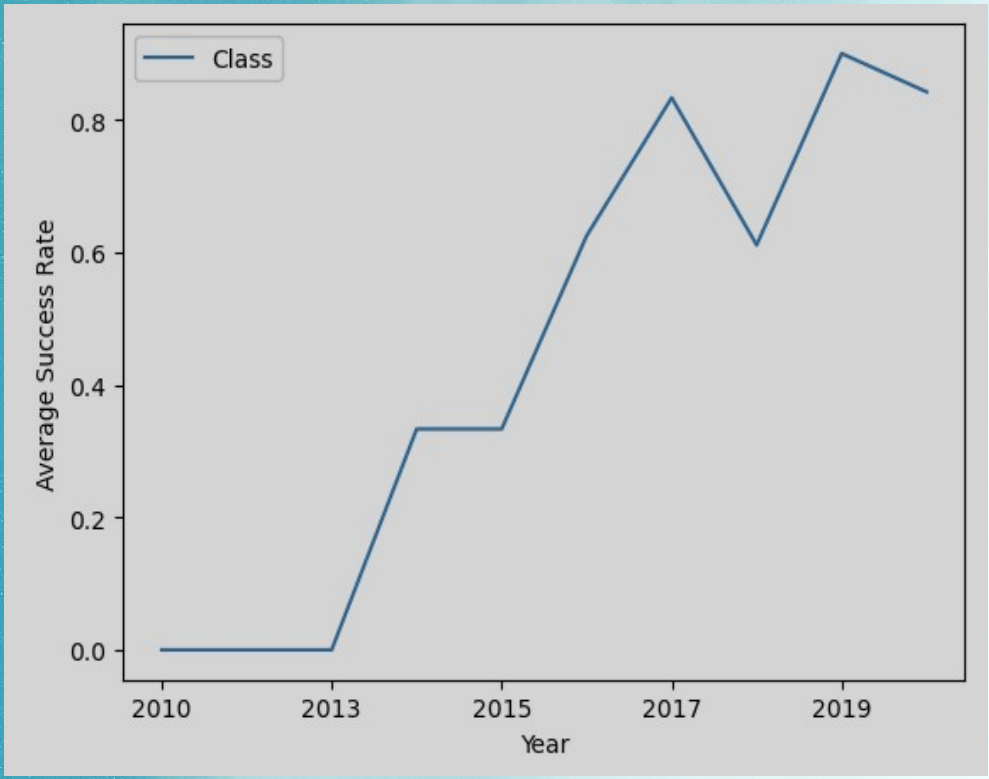
EDA Vis: *Orbit and Payload Analysis*

2. f. Annual Launch Success Trend.

The yearly average success rate shows how launch outcomes have changed over time.

This trend helps identify improvements or declines across years and assess the overall evolution of launch reliability.

GRAPH



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EDA with SQL



LAB

EDA with SQL

Objective.

Explore launch records and identify patterns related to first-stage landing outcomes using SQL.

Analysis Methods.

- Identify unique launch sites and filter launch records.
- Calculate total, average and maximum payload mass.
- Examine successful and failed landing outcomes.
- Investigate landing failures during 2015.
- Rank landing outcomes by frequency within a specified date range.

Tools.

SQL, SQLite and Pandas.

Purpose.

Extract relevant information from launch records, understand historical landing performance and support the analysis of factors associated with landing outcomes.

LAB

EDA with SQL : Query Results

1. All Launch Site Names.
2. Launch Site Names Beginning with 'CCA'.
3. Total Payload Mass.
4. Average Payload Mass by F9 v1.1.
5. First Successful Ground Landing Date.
6. Successful Drone Ship Landing with Payload between 4000 and 6000.
7. Total Number of Successful and Failure Mission Outcomes.
8. Boosters Carried Maximum Payload.
9. 2015 Launch Records.
10. Rank Landing Outcomes Between 2010-06-04 and 2017-03-20.

LAB

0201_jupyter_labs_eda_sqllite_final.ipynb

EDA SQL: *All Launch Site Names*

1. All Launch Site Names.

This query identifies all unique launch sites in the dataset.

Four distinct launch site names are returned, showing the locations used for SpaceX launches.

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

EDA SQL: *Filter with 'CCA'*

2. Launch Site Names Beginning with 'CCA'.

The query retrieves five records for launch sites whose names begin with 'CCA', identifying launch locations associated with Cape Canaveral.

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

EDA SQL: *Total Payload Mass*

3. Total Payload Mass.

The query calculates the total payload mass carried by boosters launched for NASA missions, providing an overall measure of payload mass for these launches.

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
45596

EDA SQL: *Average Payload by F9*

4. Average Payload Mass by F9 v1.1.

The query calculates the average payload mass carried by the Falcon 9 v1.1 booster version, helping characterize the typical payload mass associated with this version.

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE
WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

SQL

EDA SQL: *Successful Ground Landing*

5. First Successful Ground Landing Date.

The query identifies the earliest recorded date of a successful ground-pad landing, highlighting an important milestone in SpaceX's booster recovery program.

```
%%sql
SELECT Landing_Outcome, MIN(Date)
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	MIN(Date)
Success (ground pad)	2015-12-22

EDA SQL: *Successful Drone Ship Landing*

6. Successful Drone Ship Landing with Payload between 4000 and 6000.

The query identifies boosters that successfully landed on a drone ship while carrying payloads weighing between 4,000 and 6,000 kg, linking recovery success to a specific payload range.

```
%%sql
SELECT Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_
< 6000 AND Landing_Outcome = 'Success (drone ship)';
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

SQL

EDA SQL: *Successful and Failure Mission*

7. Total Number of Successful and Failure Mission Outcomes.

The query counts successful and failed mission outcomes, providing a summary of mission performance across the dataset.

```
%%sql
SELECT Mission_Outcome, COUNT(*)
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA SQL: *Boosters Carried Max. Payload*

8. Boosters Carried Maximum Payload.

The query identifies the booster or boosters associated with the maximum recorded payload mass, revealing the highest payload value in the dataset.

```
%%sql
SELECT Booster_Version
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EDA SQL: *2015 Launch Records*

9. 2015 Launch Records.

The query retrieves 2015 launch records involving failed drone-ship landing outcomes, including booster versions and launch site names. These records help examine unsuccessful recovery attempts during that year.

```
%%sql
SELECT substr(Date, 6,2) AS Month,
       substr(Date, 1,4) AS Year,
       Booster_Version,
       Landing_Outcome,
       Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date,0,5)='2015';
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Year	Booster_Version	Landing_Outcome	Launch_Site
01	2015	F9 v1.1 B1012	Failure (drone ship)	CCAFS LC-40
04	2015	F9 v1.1 B1015	Failure (drone ship)	CCAFS LC-40

SQL

EDA SQL: *Outcomes Between Dates*

10. Rank Landing Outcomes Between 2010-06-04 and 2017-03-20.

The query ranks landing outcomes by frequency within the specified date range, making it possible to compare the most and least common outcomes during that period.

```
%%sql
SELECT Landing_Outcome, COUNT(*)
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY COUNT(*) DESC;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	COUNT(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

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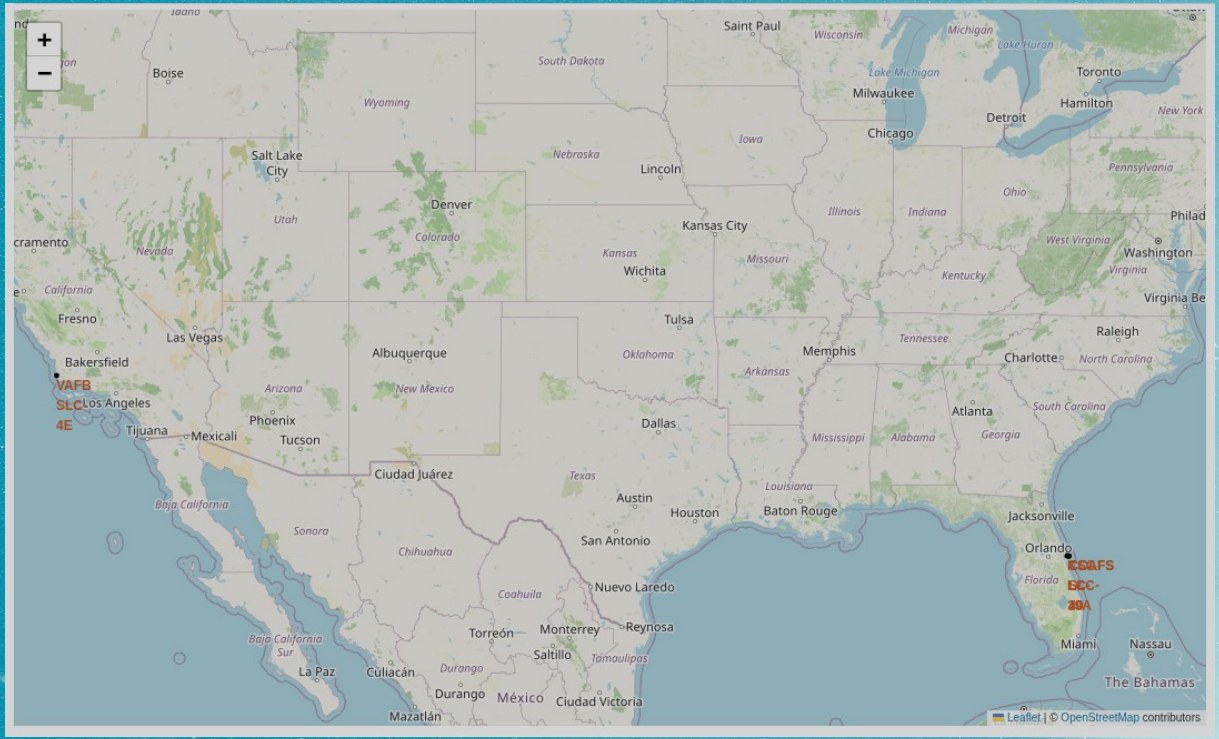
Interactive Visual Analytics



FOLIUM

Interactive Visual Analytics

Task 1.



The map shows the geographic locations of SpaceX launch sites.

Circular markers identify each site, while labels and popups display their names.

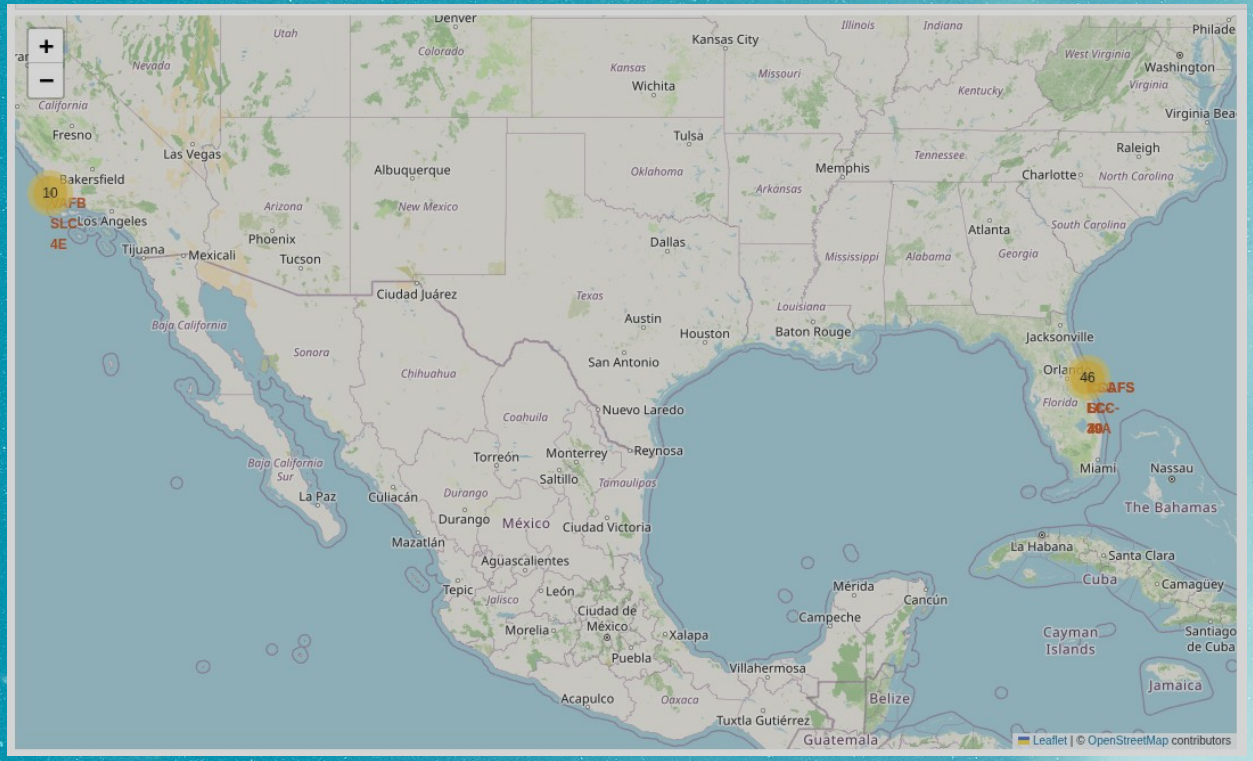
MAP

0301_lab_jupyter_launch_site_location_final.ipynb

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Interactive Visual Analytics

Task 2.



The map displays launch locations with marker colors indicating launch outcomes.

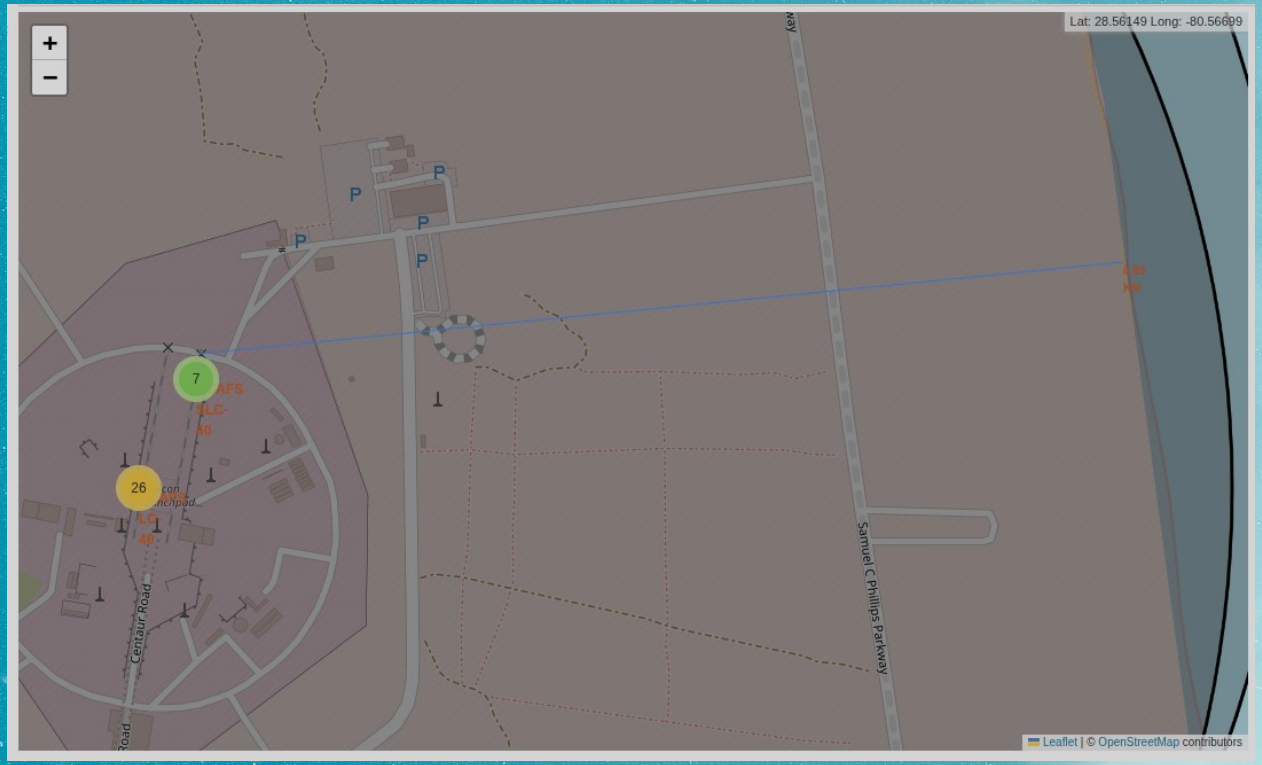
Marker clustering helps organize overlapping points and visualize the distribution of successful and unsuccessful launches.

MAP

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Interactive Visual Analytics

Task 3.



The map highlights nearby infrastructure around the selected launch site, including railways, highways, and the coastline.

Calculated distances help assess the site's geographic accessibility and proximity to key features.

MAP

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Interactive Visual Analytics

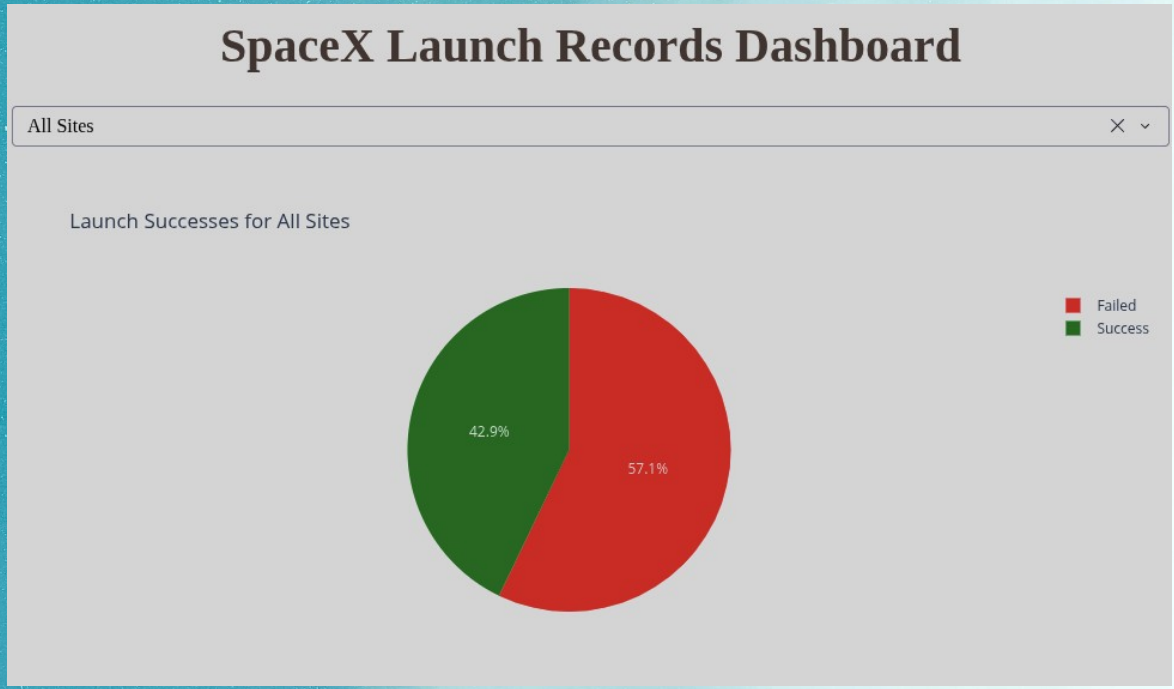


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Interactive Dashboard

Launch Success Count by Site.

The pie chart compares the number of successful launches across all launch sites. It highlights the contribution of each site to SpaceX's successful missions.



DASH

0302_spacex_dash_app.py

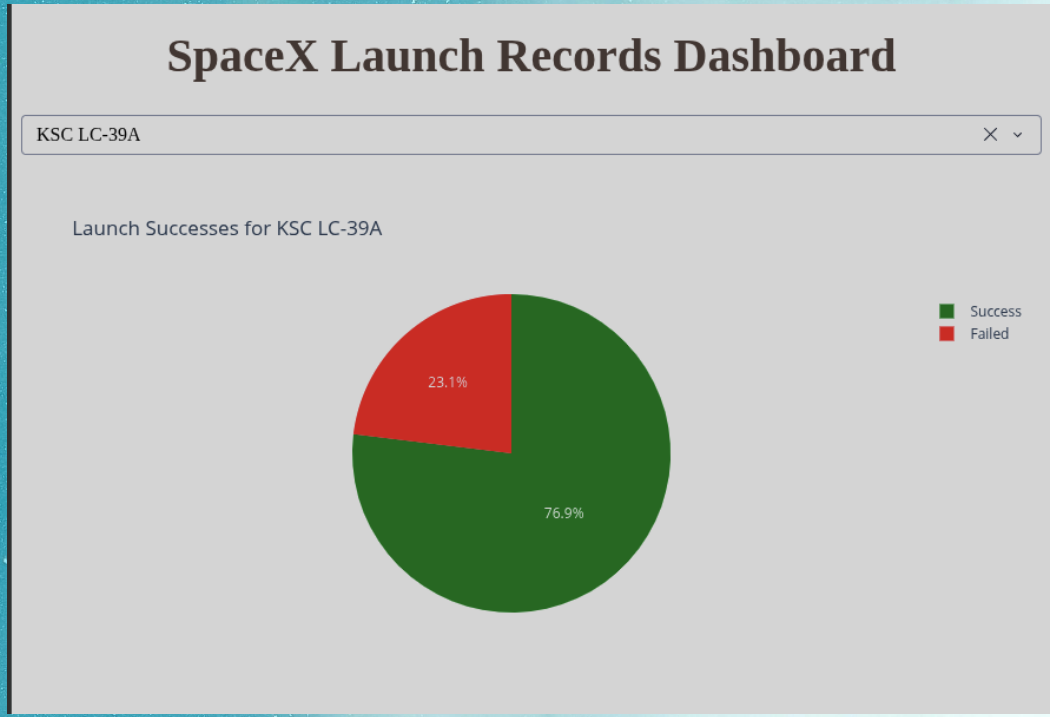
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Interactive Dashboard

Launch Success Ratio by Site.

KSC LC-39A shows the highest launch success ratio among the selected launch sites, with a rate of 76,9%.

The pie chart displays the distribution of successful and unsuccessful launches for this site.

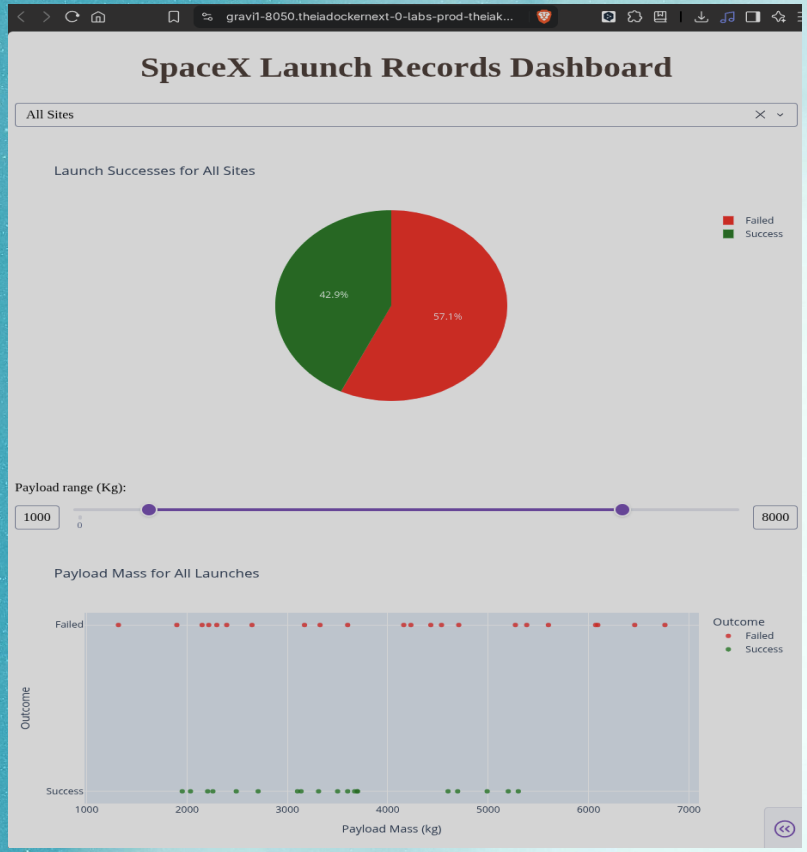


DASH

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Interactive Dashboard

Payload vs. Launch Outcome.



The scatter plot explores the relationship between payload mass and launch outcome across different launch sites. The payload range slider allows users to investigate how launch success varies across payload ranges and booster versions.

DASH

0302_5_answers.txt

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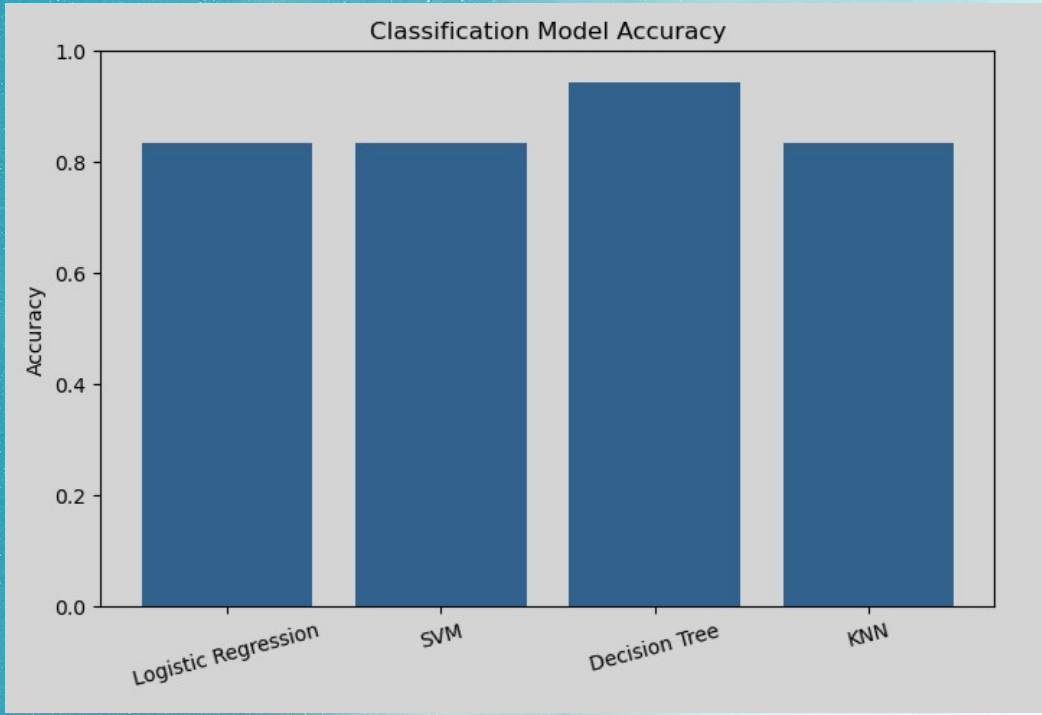
Predictive Analysis.



Predictive Analysis.

Best model.

The Decision Tree model achieved the highest classification accuracy at 94.44%, outperforming Logistic Regression, SVM, and KNN, which each achieved 83.33%. It exceeded the other models by 11.11 percentage points, making it the best-performing model in this comparison.



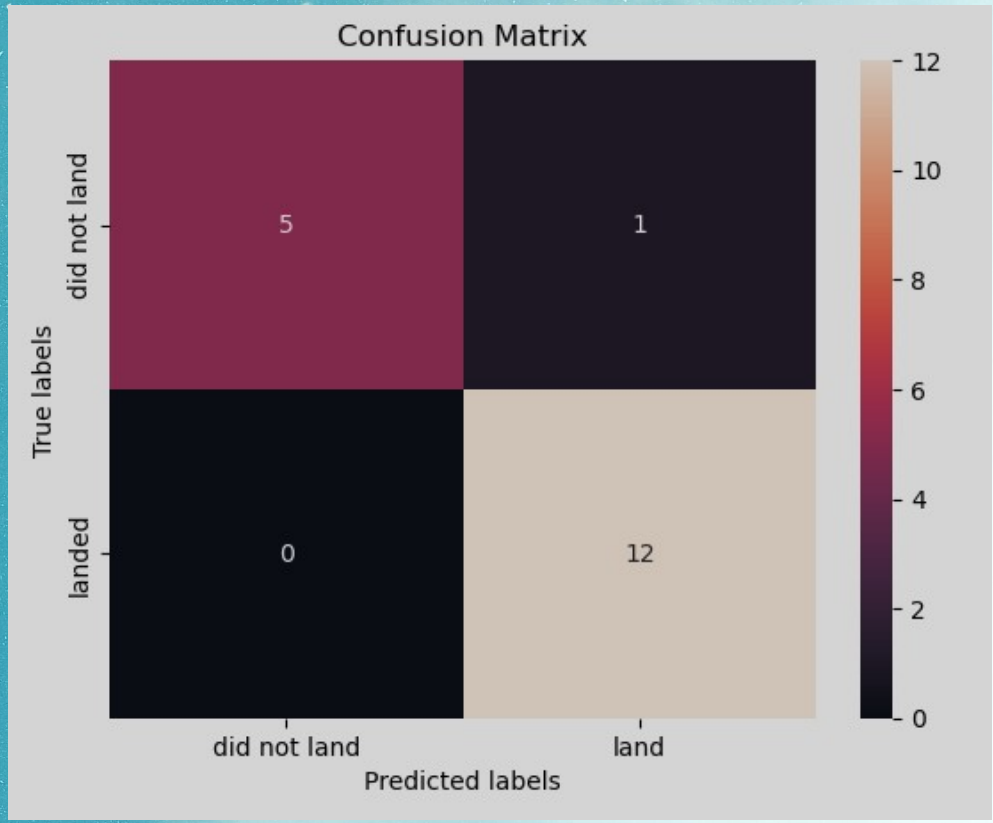
PRE

0401_jupiter_labs_spacex_ml_prediction_final.ipynb

Predictive Analysis.

Decision Tree Confusion Matrix.

The Decision Tree confusion matrix shows that the model correctly classified 17 out of 18 cases. It identified 5 cases in one class and 12 in the other correctly, with 1 false positive and no false negatives. Overall, the model achieved 94.44% accuracy, demonstrating strong classification performance.



Conclusions

Launch site performance: Launch success rates vary across sites, highlighting differences in their historical performance.

Payload and launch outcomes: Exploratory analysis helped investigate relationships between payload mass, orbit type, booster versions, and launch success.

Predictive analysis: The Decision Tree achieved the highest classification accuracy (94.44%), outperforming Logistic Regression, SVM, and KNN (83.33% each).

Overall, combining data collection, exploratory analysis, interactive visualization, and classification models provided useful insights into SpaceX launch operations and landing outcomes.

Appendix

Supporting materials for the SpaceX Falcon 9 Launch Analysis:

- SpaceX API data collection notebook.
- Web scraping notebook.
- Data wrangling notebook.
- Exploratory Data Analysis with visualization.
- Exploratory Data Analysis with SQL.
- Interactive maps created with Folium.
- Plotly Dash interactive dashboard and supporting Python code.
- Predictive analysis notebook, classification model results, and confusion matrix.

Source code, notebooks, and supporting files will be available in the project GitHub repository.

GitHub Repository:

https://github.com/loungestudio/IBM-Data-Science/blob/main/data_science/applied_data_science_capstone/

Thank you!

